Data efficiency and probability calibration of deep jet-tagging architectures on the 14 TeV top-quark benchmark

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Abstract

Top tagging — identifying the hadronic decays of highly boosted top quarks — is a standard benchmark for machine learning in collider physics. We compare four classifiers on the public Top-Quark Tagging Reference Dataset of 14 TeV proton–proton collisions: a gradient-boosted decision tree (BDT) on engineered jet-substructure variables, a convolutional neural network (CNN) on jet images, the ParticleNet graph network, and the Particle Transformer. Rather than chasing a new record area under the curve (AUC), we study two properties of direct relevance to downstream analyses: *data efficiency* and *probability calibration*. Using a fully reproducible pipeline trained on a single free cloud GPU, we sweep the training-set size from 5×10^3 to 5×10^4 jets and measure test AUC, background rejection at fixed signal efficiency, the Expected Calibration Error (ECE), and the effect of temperature scaling. The Particle Transformer attains the best discrimination (AUC = 0.978), reproducing the published ordering, but the data-efficiency sweep reveals a clear crossover: the BDT is strongest at the smallest training size and is only overtaken by the deep models as data grows. The calibration study shows that ranking power and calibration are decoupled — the jet-image CNN is the worst-calibrated model (ECE = 0.078) despite a mid-pack ranking, whereas the top-ranked transformer is comparatively well-calibrated, and a single-parameter temperature rescaling corrects every model except the CNN. The complete pipeline, trained models and documentation are released as open

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source.

Keywords: top tagging, jet substructure, deep learning, ParticleNet, Particle Transformer, data efficiency, probability calibration

1. Introduction

A boosted hadronically decaying top quark, $t \to Wb \to q\bar{q}'b$, deposits all three of its decay quarks inside a single large-radius jet that carries a characteristic three-pronged energy pattern and a mass near $m_t \simeq 172.76\,\text{GeV}$ [1]. Separating these "top jets" from the vastly more common jets initiated by light quarks and gluons (QCD jets) — *top tagging* — is one of the canonical classification problems of LHC physics and a proving ground for machine learning on high-dimensional particle data [2, 3].

The state of the art is well established. Engineered observables such as the jet mass, N-subjettiness [4] and energy-correlation functions [5] fed to boosted decision trees gave way to deep models acting directly on the constituents: convolutional networks on jet images [6, 7], the ParticleNet graph network [8], and the attention-based Particle Transformer [9]. On the standard reference dataset [10] a community comparison [11] established the now-canonical ordering with AUC values above 0.98.

What this literature rarely reports is the *cost* of those headline numbers — how much training data each architecture needs — and the *calibration* of the scores, i.e. whether they can be read as probabilities in a downstream likelihood fit. Generating labelled Monte-Carlo samples is computationally expensive, and a high AUC certifies only that a model *ranks* signal above background, not that its score is a trustworthy probability. This paper addresses both questions on the public reference dataset using a single reproducible pipeline that runs on free hardware. Our contributions are:

1. a unified, open-source pipeline that trains and evaluates the BDT, CNN, ParticleNet and Particle Transformer under identical preprocessing, splits and seeds;
2. a data-efficiency study across a decade of training-set sizes, revealing a crossover between the BDT and the deep models; and
3. a probability-calibration study showing that ranking and calibration are decoupled, with practical recommendations for downstream use.

2. Dataset and preprocessing

We use the public Top-Quark Tagging Reference Dataset [10, 11], 2×10^6 jets from 14 TeV pp collisions generated with Pythia 8 [12] and a Delphes fast detector simulation [13], clustered with the anti-k_t algorithm [14, 15] at $R = 0.8$ and $550 < p_T < 650$ GeV. For tractability we draw stratified subsamples of up to 5×10^4 jets per split and use the leading 100 constituents per jet.

For each constituent we form the feature vector $(p_T, \eta, \phi, E, \Delta\eta, \Delta\phi, \log p_T)$ relative to the p_T-weighted jet axis, standardised on the training split. For the BDT we compute thirteen engineered observables (jet mass, p_T, η, multiplicity, width, leading/sub-leading p_T fractions, τ_1, τ_2, τ_3, τ_{21}, τ_{32} [4], and C_2 [5, 16]); their distributions (Fig. 1) behave exactly as physics expects — the top mass peaks at m_t and τ_{32} separates the three-prong signal. For the CNN we bin constituents into $40\times40\times3$ jet images [6]; the averaged images (Fig. 2) show the three-prong topology in the signal$-$background difference. For the Particle Transformer we add four pairwise features $(\ln\Delta R, \ln k_T, z, \Delta\eta)$ following **(author?)** [9].

3. Models and training

The four models are: (i) a XGBoost BDT [17] on the thirteen engineered features; (ii) a ResNet-style CNN [18] on the jet images; (iii) ParticleNet [8], an EdgeConv graph network [19] on the permutation-invariant particle cloud; and (iv) the Particle Transformer [9, 20] with pairwise-feature attention bias. Related set, graph and equivariant approaches [21, 22, 23, 24] place these in context.

Deep models minimise binary cross-entropy with Adam/AdamW [25, 26], gradient clipping, a plateau scheduler and early stopping on validation AUC, implemented in PyTorch [27] with NumPy [28], scikit-learn [29] and Matplotlib [30]. Training used a free Google Colaboratory T4 GPU; the pipeline checkpoints every epoch and resumes automatically. We report AUC, accuracy, and background rejection $1/\varepsilon_B$ at signal efficiencies $\varepsilon_S = 0.5, 0.3$, with bootstrap uncertainties.

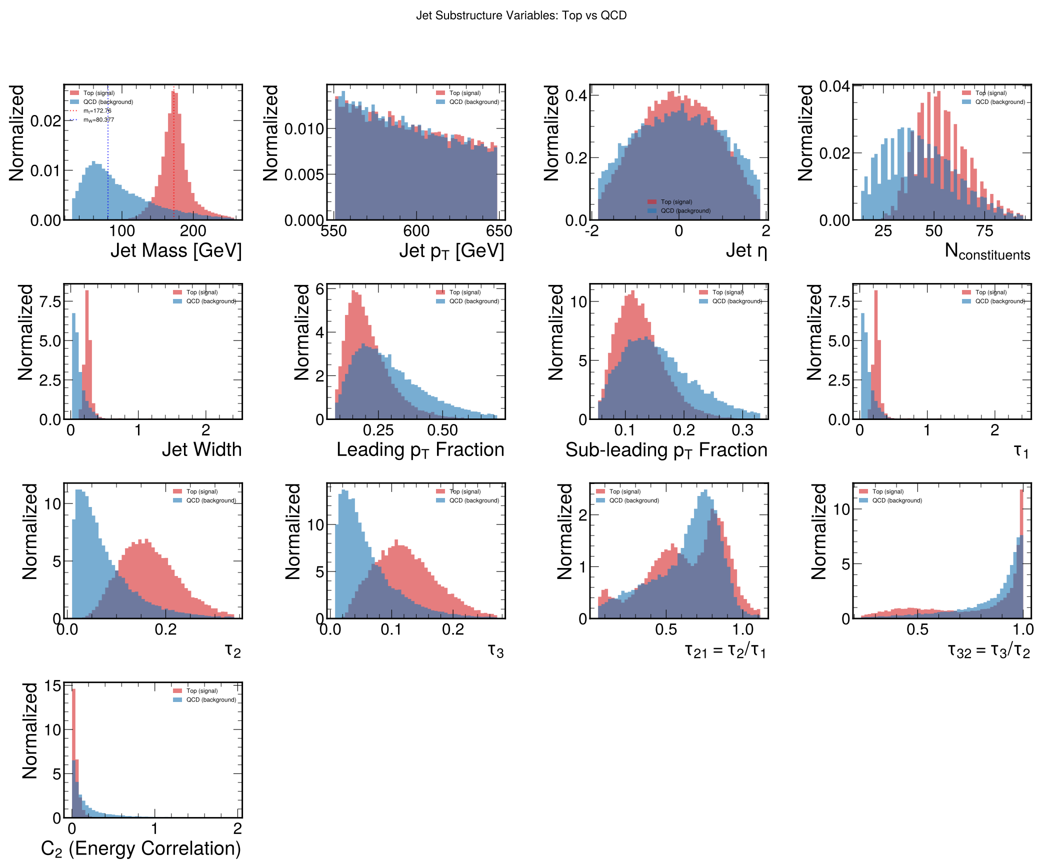

Figure 1: Engineered jet-substructure observables for top (red) and QCD (blue) jets on the test set.

4. Results

4.1. Headline benchmark

Table 1 and Fig. 3 report the headline performance. The ordering Particle Transformer (0.978) > ParticleNet (0.976) > CNN (0.972) > BDT (0.970) reproduces the community benchmark [11]; absolute values are slightly below the published saturation values because we train on a subset rather than the full 1.2×10^6 jets.

4.2. Data efficiency

Re-training each model at $N_{\text{train}} \in \{5, 10, 25, 50\} \times 10^3$ jets yields the learning curves of Fig. 4. The BDT is the most data-efficient (AUC 0.961 →

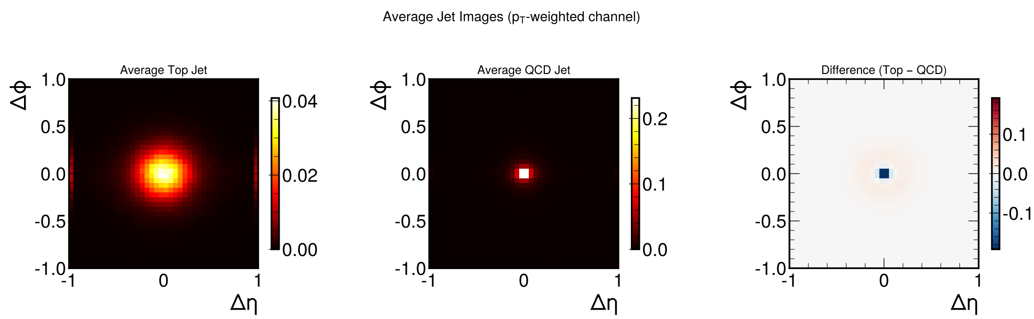

Figure 2: Averaged p_T-weighted jet images: top (left), QCD (centre) and their difference (right).

Table 1: Headline test-set performance (5×10^4-jet training run).

Model	AUC	Acc.	$1/\varepsilon_B^{0.5}$	$1/\varepsilon_B^{0.3}$
BDT (XGBoost)	0.970	0.911	93	287
CNN (jet images)	0.972	0.915	111	413
ParticleNet	0.976	0.920	130	413
Particle Transformer	0.978	0.925	136	476

0.968 across the range), whereas the deep models gain strongly and *cross over*: the Particle Transformer is the weakest model at 5×10^3 jets (AUC 0.949) but the strongest at 5×10^4 (0.974). Under a tight Monte-Carlo budget the simple, interpretable BDT is therefore competitive or preferable; the attention model only realises its advantage with sufficient data, consistent with its near-10^6-jet saturation [9].

4.3. Probability calibration

Following **(author?)** [31] we compute the Expected Calibration Error (ECE) over 15 score bins and fit a single temperature T on the validation set. Figure 5 and Table 2 show that all models are mildly over-confident ($T > 1$) but to very different degrees. Strikingly, ranking and calibration are *decoupled*: the jet-image CNN is the worst-calibrated by an order of magnitude (ECE $= 0.078$) despite ranking third on AUC, whereas the top-ranked Particle Transformer is comparatively well-calibrated (ECE $= 0.012$). Temperature scaling brings the BDT, ParticleNet and transformer to ECE $\lesssim$ 0.008 but only halves the CNN's error (to 0.064): the CNN's miscalibration is

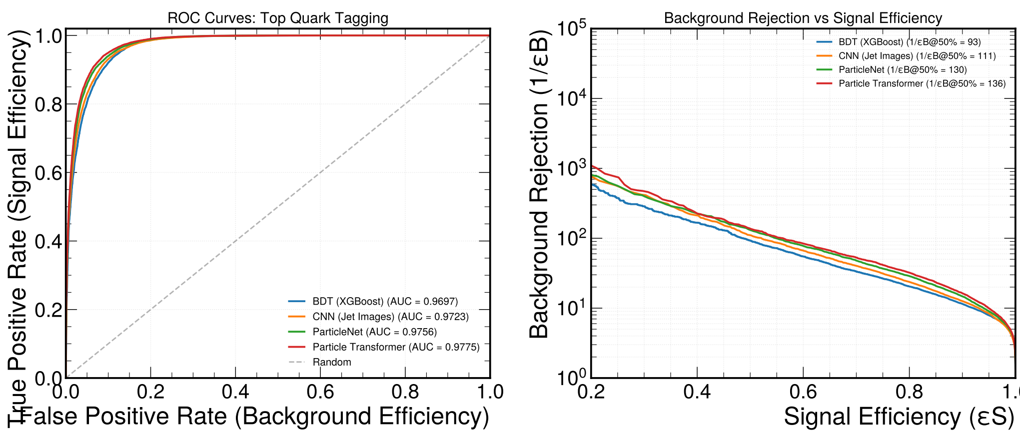

Figure 3: ROC curves (left) and background rejection vs. signal efficiency (right).

Table 2: Calibration at the 5×10^4-jet endpoint: ECE, fitted temperature T, and ECE after rescaling.

Model	ECE	T	ECE_T
BDT (XGBoost)	0.007	1.04	0.006
CNN (jet images)	0.078	1.54	0.064
ParticleNet	0.008	1.05	0.008
Particle Transformer	0.012	1.12	0.007

structural and would require a richer recalibration such as isotonic regression [32].

4.4. Interpretability

The class-token attention of the Particle Transformer (Fig. 6) concentrates at small angular distance ΔR for QCD jets (a single core) but spreads to large ΔR for top jets, reflecting the broader multi-prong energy flow of the top decay. The BDT feature importances are dominated by the jet mass, τ_{32} and multiplicity, the physically expected discriminators.

5. Conclusions

We have presented a reproducible comparison of four top-tagging architectures focused on data efficiency and probability calibration. The literature

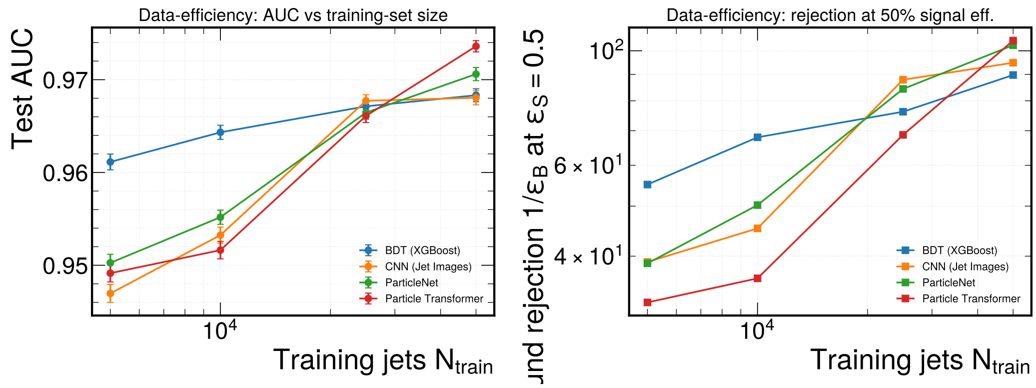

Figure 4: Data-efficiency curves: test AUC (left) and background rejection at $\varepsilon_S = 0.5$ (right) vs. training-set size.

ordering is reproduced, but two practical messages emerge. First, data efficiency separates the models: the BDT is best in the small-data regime, with the deep models overtaking it only as data grows. Second, ranking and calibration are decoupled — the best-ranked transformer is well-calibrated, while the CNN, mid-pack on AUC, is the worst-calibrated and is not fully corrected by temperature scaling. We recommend reporting temperature-scaled probabilities by default and treating CNN scores with particular caution. Natural extensions are to scale the sweep to the full dataset, to study robustness under pile-up, and to compare non-parametric recalibration.

Code and data availability

The complete pipeline, trained models, and documents are openly available at `https://github.com/ramc77/fyp-jet-tagging`. The dataset is the public Top-Quark Tagging Reference Dataset [10] (`https://zenodo.org/records/2603256`).

Acknowledgements

We thank the maintainers of the Top-Quark Tagging Reference Dataset and the open-source scientific-Python and Google Colaboratory ecosystems.

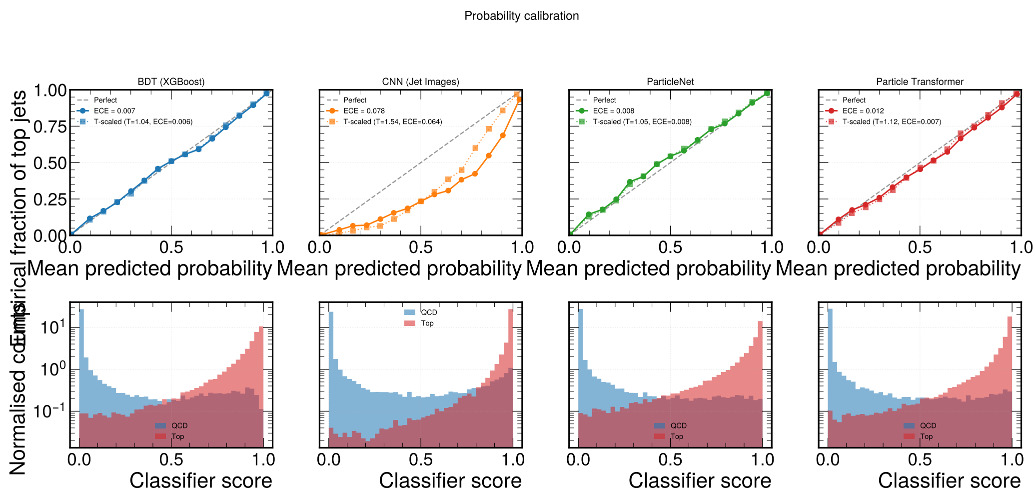

Figure 5: Reliability diagrams (top) and score histograms (bottom). Solid: raw scores; dotted: temperature-scaled. The CNN deviates most.

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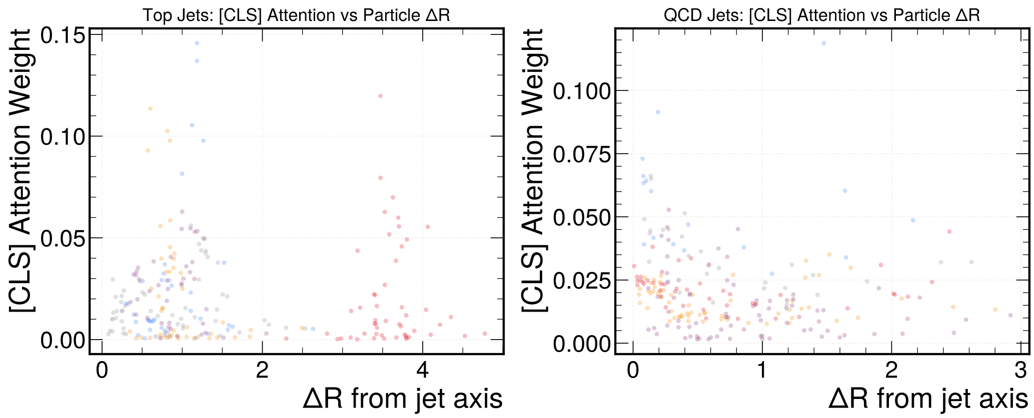

Figure 6: Class-token attention vs. constituent ΔR for top jets (left) and QCD jets (right).

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